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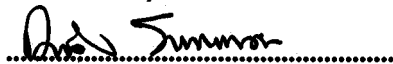
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METHODOLOGY FOR THE DEMONSTRATION OF COMPLIANCE WITH SINGLE RAIL REED TRACK CIRCUITS ON THE A.C. RAILWAY

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IMPLEMENTATION

The provisions of this code of practice may be used from the date of issue.

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I. OBJECTIVES

I.1 Purpose

The purpose of this document is to provide a methodology to demonstrate compatibility with 'RT' type single rail track circuits on the a.c. railway on Railtrack controlled Infrastructure. This is achieved by:

- Providing a characterisation of the reed track circuit receiver.
- Providing a characterisation of the infrastructure.
- Providing a methodology to define the susceptibility of a given reed track circuit.

Combining these elements will enable a full characterisation of a reed track circuit applicable to Railtrack infrastructure. This characterisation will take the form of a permissible train current which can then be compared to the footprint of a piece of traction or rolling stock to enable a case to be made. Methods are included to analyse intact and faulty infrastructure.

I.2 Scope

This document covers RT type reed track circuits installed in single rail mode on a.c. electrified Railtrack controlled infrastructure. This also encompasses the older type 'R' equipment when installed in single rail mode where these are not fitted with step up transformers.

I.3 Installation types covered by this Document

This document covers reed track circuits installed in single rail mode on the a.c. railway. It does not cover specific double rail installations that may be present within tunnels etc. Double rail installations will require a specific assessment based on the configuration as installed. Installations on d.c. electrified lines are generally installed in double rail configuration including impedance bonds and are therefore not covered in this document.

There may be instances of type 'R' reed track circuits that have been installed with step-up transformers fitted to the receiver, these track circuits are also excluded from the scope of this document.

The analysis here assumes that the a.c. electrification is installed as a 'conventional' system and is valid for rail return, rail and return conductor return and booster transformer schemes. Autotransformer fed sections of railway are not covered by this document and will require a separate analysis and/or a reissue of this document.

1.4 Equipment types included by this Document

The equipment covered within this document is the type 'RT' reed track circuit equipment manufactured by GEC (now ALSTOM). This equipment has an adjustable track filter that affects the sensitivity of the receiver. For the analysis presented here the filter is assumed to be set at its most sensitive setting. At this setting the RT type reed receiver is more sensitive than the earlier type 'R' reed track circuit equipment. Due to this difference in susceptibility characteristic type 'R' track circuit equipment is encompassed within the analysis presented here.

This document does not cover application specific receivers used in double rail track circuits on the a.c. railway nor does it cover more recently developed high power double rail reed track circuits that operate with higher rail to rail voltages than the type 'RT'.

2. SOURCE DOCUMENTS

2.1 Group Standards

1. GK/RC0752 General information on track circuits, Issue 2, December 1998.
2. GK/RT0252 Minimum requirements for the bonding of track circuits, Issue 2, February 1996.
3. GK/RC0763 Reed Track Circuits, Type RT Including High Performance Arrangement, Issue 1, December 1999.
4. Signalling Installation Handbook, GS/IH0001, Issue 2, Railtrack S&SD, January 1994.

2.2 Previous Safety Case documentation

5. "Results of Single Rail Reed Track Circuit Tests at the Old Dalby Test Site", AEAT Rail, I20-dr0394, version a0, dated 27th July 1998.
6. "Measurement and Characterisation of Reed Track Circuit Receiver" AEAT Rail, I20-dr0399, version a0, dated 19th June 1998.
7. "Determination of Permissible Rail/Wheel Interface Current Levels for Single Rail Reed Track Circuits on the AC Railway", AEAT Rail ref. I20-dr0397, version b0, dated May 1999.
8. "Juniper Project - Reed Track Circuit Interference Limits - AC Routes", AEAT Rail ref. PRJ-SCSE(JUN)-RP-20, Issue 3, dated July 1999.
9. "Track parameters for Interference studies" N Rambukwella, AEAT Rail Report number STG-D-EPS(373)RP-18, Issue 1, January 1995.
10. "Reed Track circuit parameters" fax from D. M. Poole to N Rambukwella dated 12/5/93.

11. “Type RT Reed Track Circuit Equipment Part I Standard Power Jointed Track Circuits” IB147 Part I Issue 4, Alstom Signalling, October 1991
12. “Suitability Report - Circuit Equations for Reed Track Circuit Installed on the ac Railway” TCI, I20-dr0396, version a0, dated 11th November 1998.
13. Class 92 Locomotive EMC Safety Assessment Volume 3 – Single Rail Reed Track Circuits Compatibility with Single Rail Reed Track Circuits on the 25kV ac Railway”, AEA Technology Rail, ref. I20-dr0454 , version b0, dated May 1999.

2.3 Other standards

14. prEN 50238 ‘Railway Applications: Compatibility between Rolling Stock and Train Detection Systems’ Final draft 1998.

2.4 Railtrack IDI documentation

15. “Reed Receiver Test Report - Railtrack Industry Data Initiative”, AEAT Rail, ESID9079, version 2, March 2000.

2.5 Other Documents

16. “Railway Safety - HM Chief Inspecting Officer of Railways’ Annual Report on the safety record of the railways in Great Britain during 1995/6” HSE Books

3. INFRASTRUCTURE ASSUMPTIONS

The infrastructure assumed during the analysis is as described below, further assumptions are listed in Appendix B:

- Track circuits are installed in single rail mode without the use of impedance bonds.
- The cross bond spacing is assumed to be 750m. The group standard (ref.1) states that at least 1 yellow bond per km is required, preferably adjacent to feed or relay connections. There may be additional cross bonds within the track circuit but these will not affect the analysis presented since they do not affect the impedance of the effective traction return rail as seen from the ends of the track circuit.
- The rail impedance at reed frequencies is as quoted in ref. 7, i.e. 346 mΩ/km and 0.743 mH/km for a frequency of 400Hz. This is taken from ref. 9 and is

representative of flat bottomed BSI 13A rails. The expected variation in this value is in the order of 1% over the frequency range of interest 363 - 423 Hz.

- The Signalling Installation Handbook (ref. 4) states that track circuit lengths can be up to 1200m, this is used as the worst case installed track circuit.
- Track filter primary impedance: 2.8Ω resistive (refs. 7 and 10). This represents the impedance presented by the primary of the track filter at resonance with the highest sensitivity selected.
- All of the traction current returns to source via the rails. Booster transformers and return conductors are assumed not to exist in the analysis presented. This generally leads to pessimistic results for electrification schemes where booster transformers and return conductors are used. This configuration is the most common arrangement on Railtrack infrastructure. In the case of rail return only systems the results are realistic.
- The effect of earth masts connected to the traction rail are ignored. The effect of masts would be to reduce the current returning to source via the traction and signal rails. This will reduce the current flow in the signal rail and therefore it is a pessimistic assumption.
- The transmitters of track circuits on adjacent tracks are assumed to be open circuits. This is a pessimism in the analysis since these use the same track filters as the receiver and will conduct some traction current.
- The effect of the mutual coupling between the rails and the overhead has been ignored. It is considered that these mutual impedances would have a second order effect on the signal rail current. The current flow in the overhead line would induce voltages in the traction and signal rail that were equal in both magnitude and polarity and as such would have no effect on the current flow in the signal rail. The effect of the mutual coupling between the rails would have the effect of reducing the current in the signal rail and therefore this assumption is pessimistic. This has been shown in ref. 12.
- The track is assumed to be of standard gauge
- The sleepers in use are assumed to be concrete or wood, steel sleepers are not included in the analysis presented here.
- UIC 60 Rail is not included in the assessment as presented. The methodology given here is applicable but the parameters of this rail type at the frequency of interest will need to be determined and used in the appropriate calculation.

- The presence of check rails will reduce the effective impedance of the rail to which they are attached. This is not included in this document, however the methodology provided allows for the effect of such differences to be evaluated.
- A cross bond failure on a section consisting of two or more tracks essentially reduces the configuration to no worse than that of a single track railway in the absence of another infrastructure fault e.g. a broken rail. The analysis presented considers only one infrastructure fault occurring at any one time.

4. SUSCEPTIBILITY ANALYSIS

4.1 Receiver analysis

The reed track circuit is a single frequency track circuit that is usually installed in single rail mode on the a.c. railway. The frequencies of operations are: 363Hz, 366Hz, 369Hz, 372Hz, 375Hz, 378Hz, 381Hz, 384Hz, 408Hz, 417Hz and 423Hz. These frequencies are commonly referred to by their channel numbers f211 to f221 (ref. 11).

4.1.1 Receiver components

The reed track circuit receiver is composed of the following components as shown in figure 1.

- Track filter
- Electromechanical filter (the 'Reed' filter)
- Amplifier
- Follower relay

The track filter prevents large amounts of traction return current flowing through the receiver. It consists of a tuned circuit on the primary of a transformer and acts as a broad 'band pass' filter. The frequency of the pass band is adjustable. The resistance in the primary of the filter is able to be set to 2.8Ω (setting 23), 5Ω (setting 22) and 7.2Ω (setting 21) (ref.10). The lower resistance settings correspond to higher sensitivities. The track filter does not introduce any significant time delay into the system.

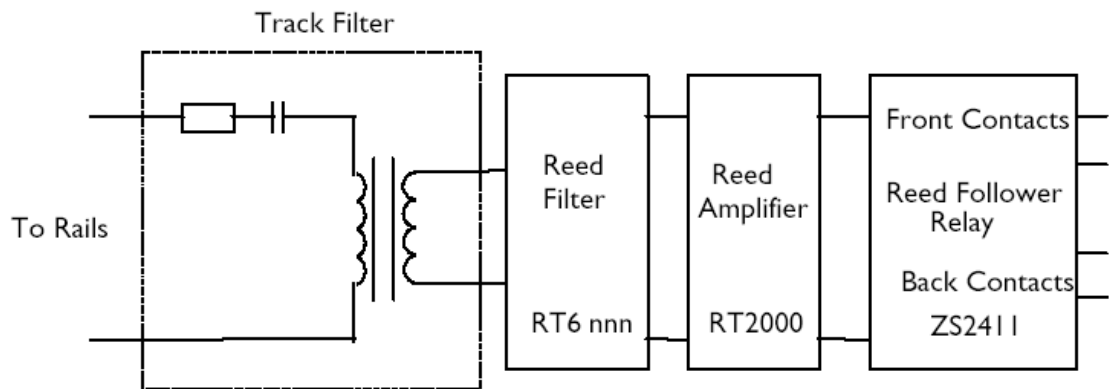
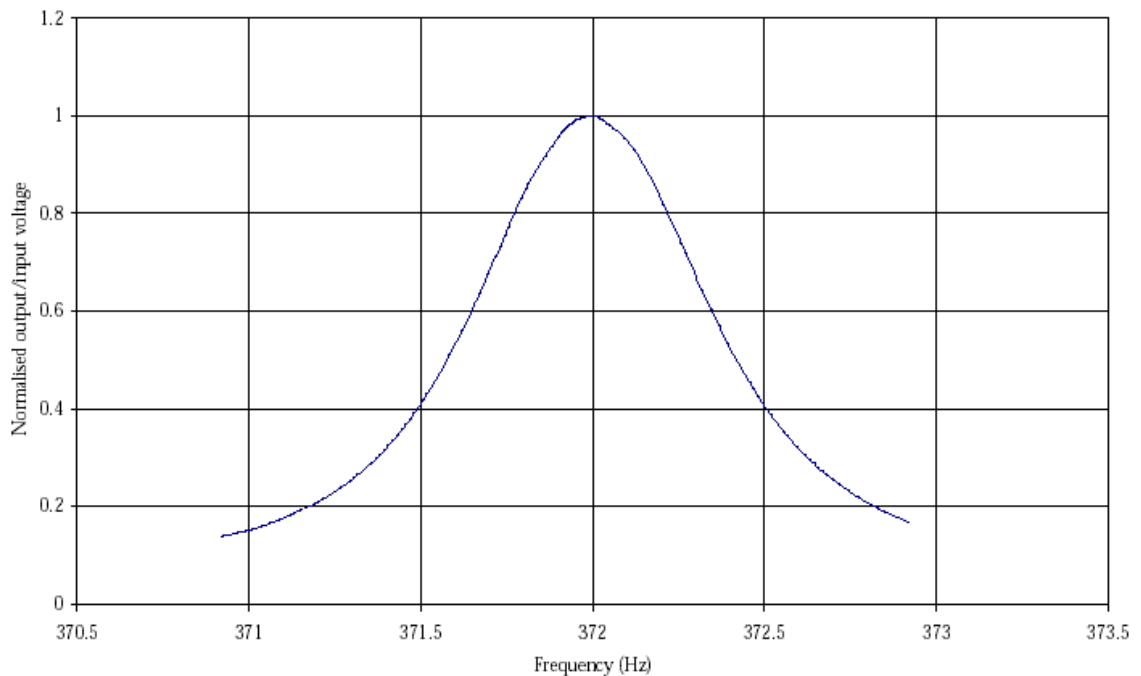


Figure 1: Constituent parts of the reed receiver
 'nnn' refers to the three digit code used for channel identification

The output from the track filter is fed into the reed filter itself that actually consists of two vibrating reeds coupled via a resilient mounting. This filter has a very sharp frequency response with a quoted bandwidth of 0.5 Hz (ref. 11). A typical reed filter characteristic is given in Figure 2.



Due to the nature of the filter there is a time delay between the application of a signal to the input and the maximum output level being reached. There is also a delay between the input signal being removed and the output decaying to zero.

The reed filter output is then amplified and applied to the follower relay. The reed amplifier also introduces a time delay into the system.

The follower relay is normally connected into the signalling system via a slow to pick track repeat relay to specification BR933 (TPR) (ref. 1). The analysis here is therefore somewhat pessimistic in ignoring this. TPR fitment is historically difficult to confirm and it is for this reason that this is not assumed in the analysis. The benefit to be gained from the fitment of a single TPR is marginal. This is demonstrated in ref. 7.

4.1.2 Analysis of receiver components

The track filter is characterised in terms of the ratio of its input current to output voltage. This quantity is found to be independent of the filter setting. The filter setting will affect the amount of interference current that will flow in the primary of the filter and thus affect the output voltage for a given infrastructure scenario.

Ratio of input current/output voltage for track filter = 1.6 mA/mV (ref. 7)

In the case of single rail track circuits the current in the primary of the track filter is that which flows in the signal rail of the track circuit.

The receiver is characterised in terms of its response to interference applied to the input of the reed filter. The characterisation is based on the time for which a signal must be applied in order that the front contacts of the follower relay to just close. This quantity is expressed as a voltage - time product enabling a time variant susceptibility to be determined.

This voltage time product: 88000 mV*ms (ref. 7)

N.B. This value is suitable for use for durations between 200ms and 1 second.

The reed receiver has been characterised for durations shorter than 200ms by testing described in ref. 15.

For durations greater than 1 second the maximum permissible reed filter input voltage is taken as the maximum 'drop away' voltage, this being 65mV (ref. 10).

Using this value the permissible reed filter voltage/time curve is as shown in figure 3.

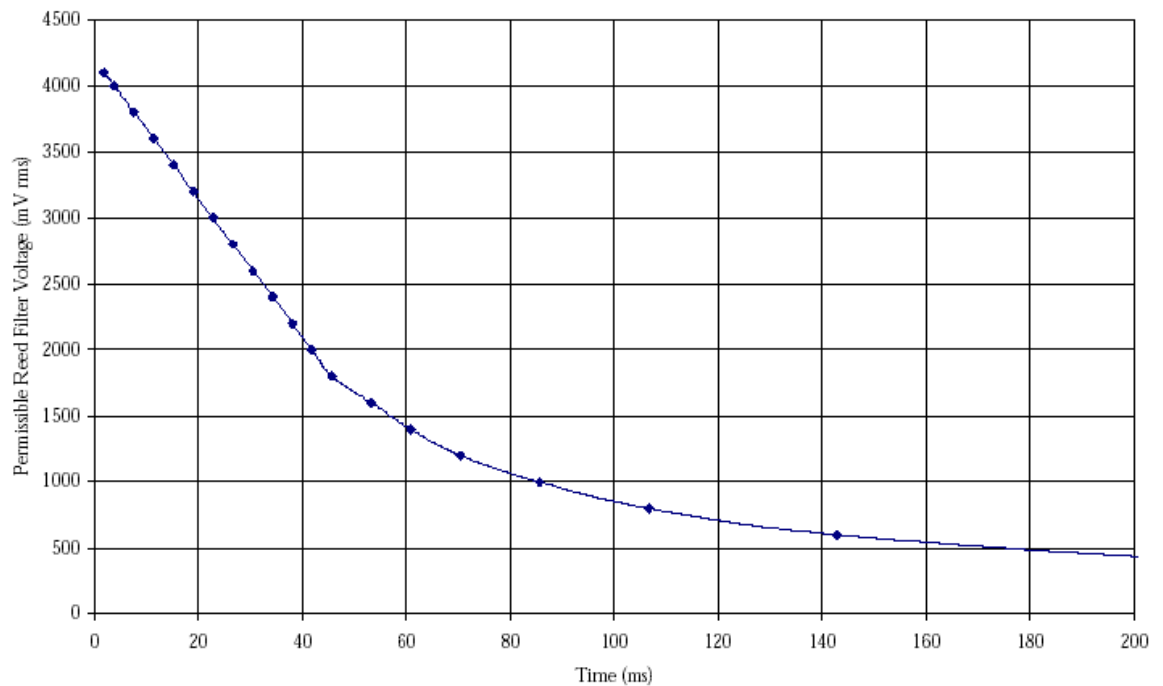


Figure 3: Permissible reed filter voltage vs time <200ms

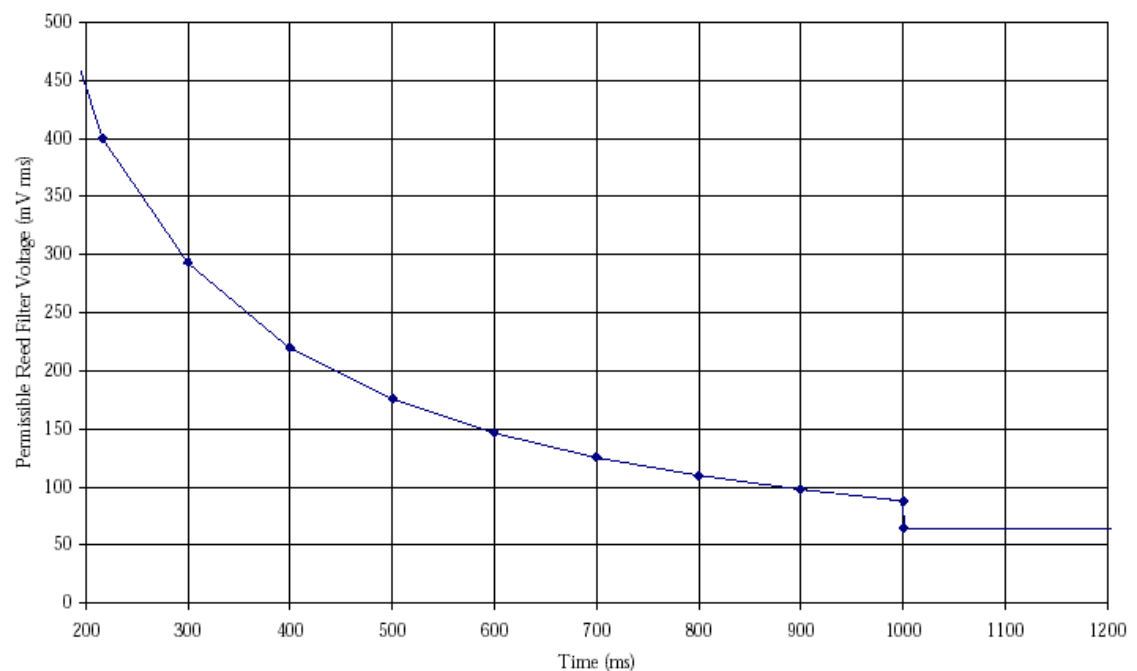


Figure 3: Permissible reed filter voltage vs time >200ms

The curve is horizontal at durations greater than 1 second since this voltage level is below that at which the receiver would 'drop'. This level is 65 mV rms. By applying the scaling factor of 1.6 mA/mV of the track filter the permissible current in the

primary of the track filter and hence the signal rail is determined. This current is shown in figure 4. Again the curve is horizontal at durations greater than 1 second (steady state), this level is 104 mA rms.

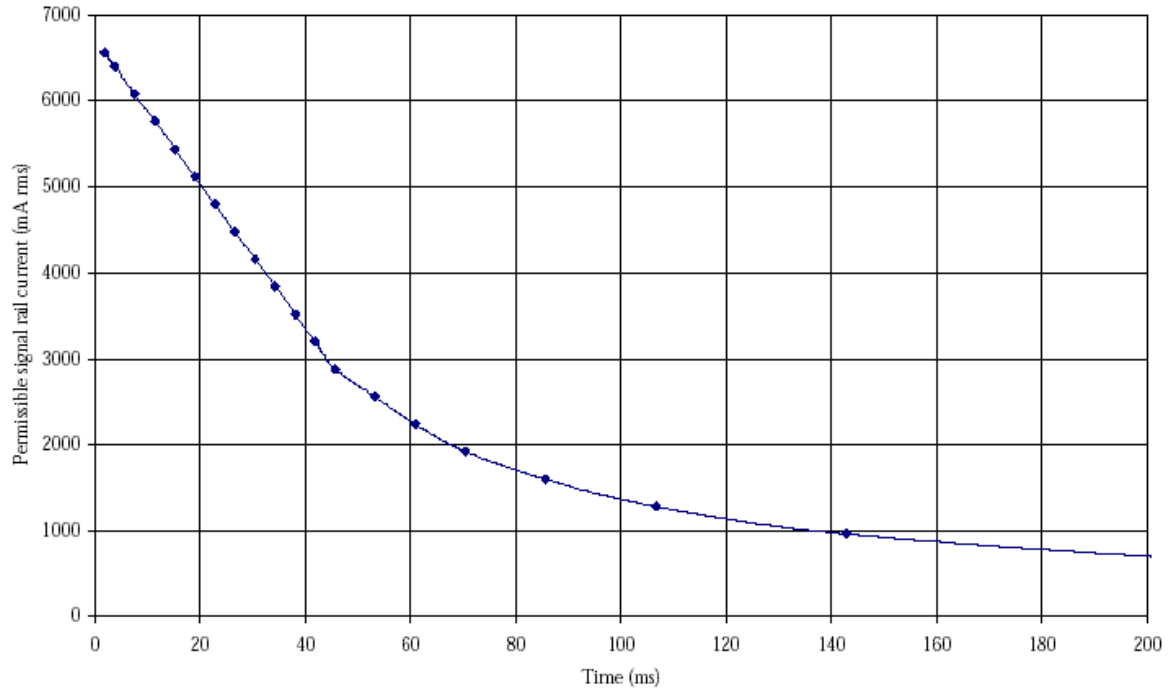


Figure 4: Permissible track filter primary current vs time <200ms

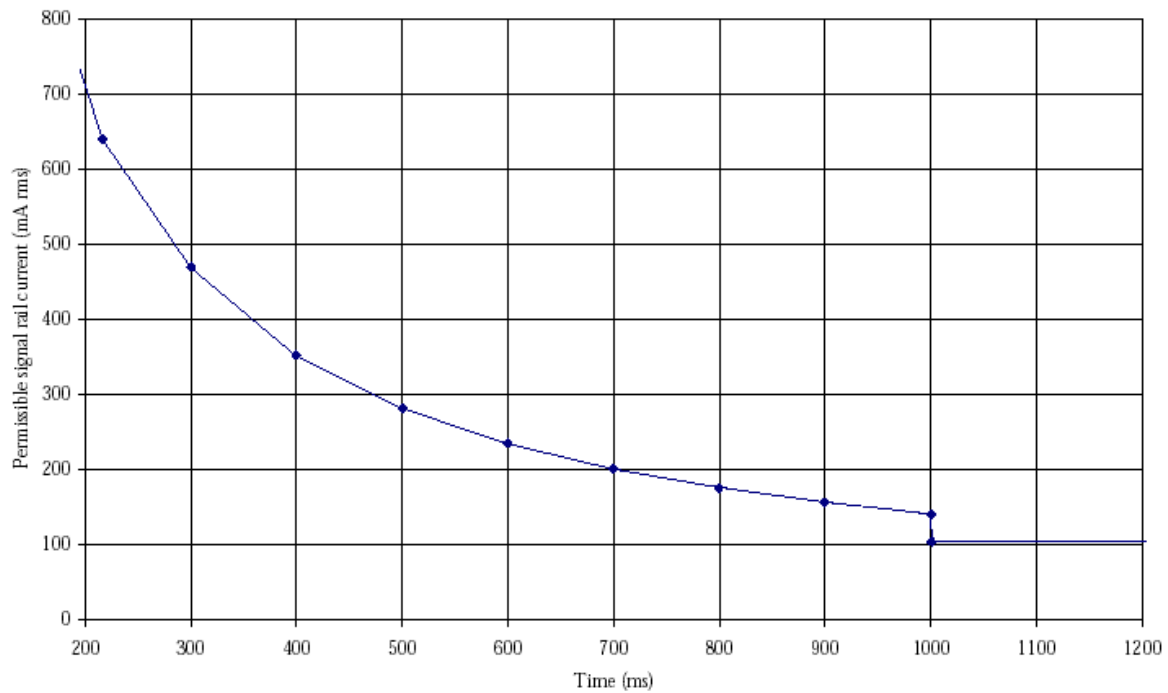


Figure 4: Permissible track filter primary current vs time >200ms

4.2 Infrastructure analysis

The infrastructure analysis allows the proportion of traction return current that will flow back to source via the signal rail of the track circuit and hence the primary of the track filter to be determined. As stated in section 3 all of the return current is assumed to flow back to source through the rails rather than any other return conductors that may exist.

The infrastructure analysis becomes simply the analysis of an impedance network formed by the rails and the reed receiver. The amount of traction generated interference current that flows down the signal rail can be expressed as a fraction of the total. This leads to the calculation of a transfer function ' Ψ ' for a particular infrastructure condition.

Where: $I_{TDS} = \Psi I_{RS}$

I_{TDS} = Current flowing in the signal rail or primary of track filter

I_{RS} = Total interference current produced by rolling stock.

$\Psi < 1$

The simplest case for this calculation is shown in figure 5.

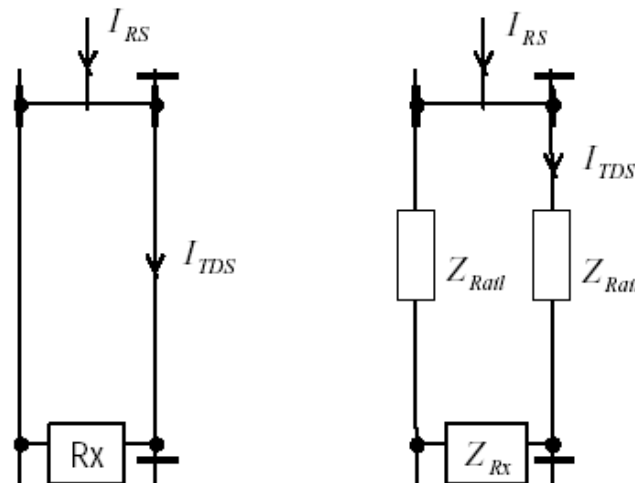


Figure 5: Simple infrastructure case for transfer function Ψ

From figure 5 it can be seen that transfer function is dependent on the length of the track circuit, the rail impedance and the impedance of the receiver. The parameters for the rail and receiver are given in section 3. The circuit is a simple current divider with one branch being the traction return rail and the other being an equal length signal rail with the extra impedance due to the receiver connected across the rails.

The infrastructure analysis can be expanded to cover other infrastructure condition such as number of tracks and infrastructure failure by changing the circuit under consideration. For multitrack scenarios the traction rail is replaced by an “effective traction rail”. This “effective traction rail” which will have a reduced impedance compared to that in the single track scenario shown in figure 5. This is due to the parallel connection of a number of traction rails via the cross bonds assumed to exist at each end of the track circuit. This scenario is shown in figure 6.

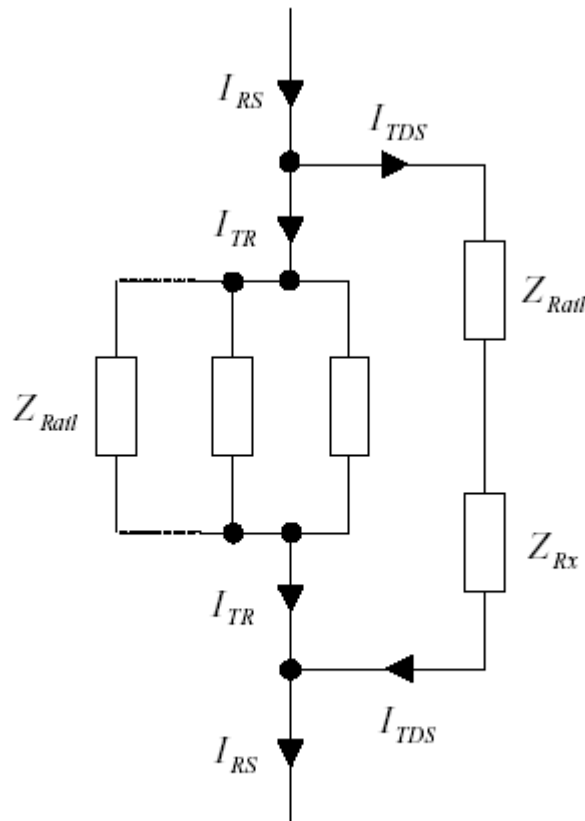


Figure 6: Circuit diagram for determining the transfer function

Figure 6 shows the impedance network to be used to determine the transfer function Ψ . This network accounts for the presence of the traction rails of adjacent tracks, one extra return rail is included per track. The terms used in figure 6 are as follows:

I_{TDS} = current in signal rail or primary of track filter

I_{RS} = Current from rolling stock

I_{TR} = Current in effective traction rail

Z_{RX} = Impedance of primary of track filter (2.8Ω resistive)

Z_{RAIL} = Impedance of the length of rail at the frequency of interest based on a resistance of $346\text{m}\Omega/\text{km}$ and inductance of 0.743 mH/km

From figure 6 the value of Ψ for a given scenario is:

$$\Psi = \frac{Z_{RAIL}/n}{Z_{RAIL}/n + (Z_{RAIL} + Z_{RX})}$$

where n is the number of tracks.

From this it can be seen that in the case of intact infrastructure the value of the transfer function Ψ reduces as the number of tracks increases. This means that a smaller proportion of the interference generated by the rolling stock flows through the primary of the track filter as the number of tracks increases.

This equation also indicates that the transfer function ψ is very dependent on the value of Z_{RAIL} which is linearly dependent on the length of the track circuit of interest and the per unit impedance of the rail in question. The value of Z_{RX} is taken as the worst case i.e. the resonant value of 2.8Ω .

4.3 Track circuit susceptibility

4.3.1 Line current limits

Once the value of the transfer function ' Ψ ' is known this can be used to determine the maximum permissible interference current that can flow in the system. The maximum permissible interference current is given by:

$$I_{RS} = I_{TDS} / \Psi$$

In this case I_{TDS} is set to the maximum permissible from the susceptibility characteristic to give the maximum permissible rolling stock current.

This equation can then be applied to the curve shown in figure 4 to give the permissible interference current. This maximum train interference current can then be apportioned down to account for the effects of background interference that will also be present in the system from other sources.

This apportionment is a reduction in the maximum permissible current to a third of that calculated. This is considered appropriate to account for differences in track circuit parameters and interference from other sources, such as transmitter breakthrough and trains within the same feeder section. The levels of interference due to these other sources is unquantifiable and is location dependent. These sources must add constructively to themselves and the train generated interference. This is unlikely due to the following reasons:

- The interference must be of the correct frequency due to the narrow band nature of

the filter.

- The interference must have the correct phase relationship to add constructively.
- The calculated permissible current will be based on the longest allowable track circuit.
- The train is assumed to run on at the transmitter end which may not always be the case.
- All of the return current is assumed to stay in the rail.

4.3.2 Application to non-rectangular shaped transients

The transient event should be characterised as a square envelope having an amplitude of 90% of the peak value of the transient and a duration of the time taken for the transient to decay to 10% of its peak value. If this characterisation sits below the derived permissible train current then no further consideration of this single transient event is required. However if the permissible train current curve is exceeded, alternative analysis would be required. For example the transfer function calculated could be used to scale a measured train current to enable this to be played into either a validated mathematical model of a reed receiver or a real receiver.

4.3.3 Axle to axle or longitudinal voltage limits

The train under consideration also poses a threat to track circuits installed on adjacent tracks in multi-track areas. Due to the flow of circulating currents within the train and the rails a voltage will be developed across the outermost axles of the train. This axle to axle voltage acts as a source to drive a current through the receiver of a track circuit on an adjacent track by virtue of the cross-bonding between the tracks. This may have an effect on the adjacent track circuit if the current caused to flow due to the axle to axle voltage is high enough to reach the drop away level of the track circuit. If this is the case the track circuit may fail to drop when occupied by a train.

This scenario is depicted in figure 7.

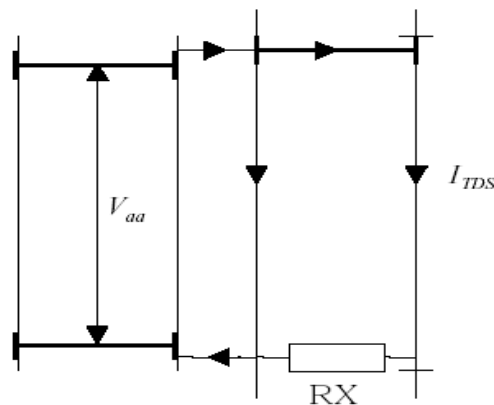


Figure 7: Circuit diagram to consider axle to axle voltage

In figure 7 the axle to axle voltage is denoted by V_{aa} and the current in the signal rail is I_{TDS} . From figure 4 the permissible signal rail current is 104 mA rms for 'steady state' conditions. The worst case condition for axle to axle voltage occurs when the train is the same length as the cross bond spacing. If the rail impedance is ignored, the axle to axle voltage that is required to drive 104 mA rms through the primary of the track filter (2.8Ω) would be 296.8 mV rms.

This value represents the absolute maximum voltage that can appear across the cross bonds and needs to be apportioned down to account for the presence of background and other sources of interference. If the same factor of 3 is used this is reduced to 99 mV rms for the axle to axle voltage of the train under consideration.

5. WORKED EXAMPLE - NORMAL INFRASTRUCTURE

5.1 Infrastructure considered - (track circuit handbook)

For the example presented here the maximum installed track circuit length of 1200m will be analysed as installed on single and double track sections of railway.

5.1.1 Infrastructure analysis

Schematics of the infrastructure considered in this analysis are shown in figure 8.

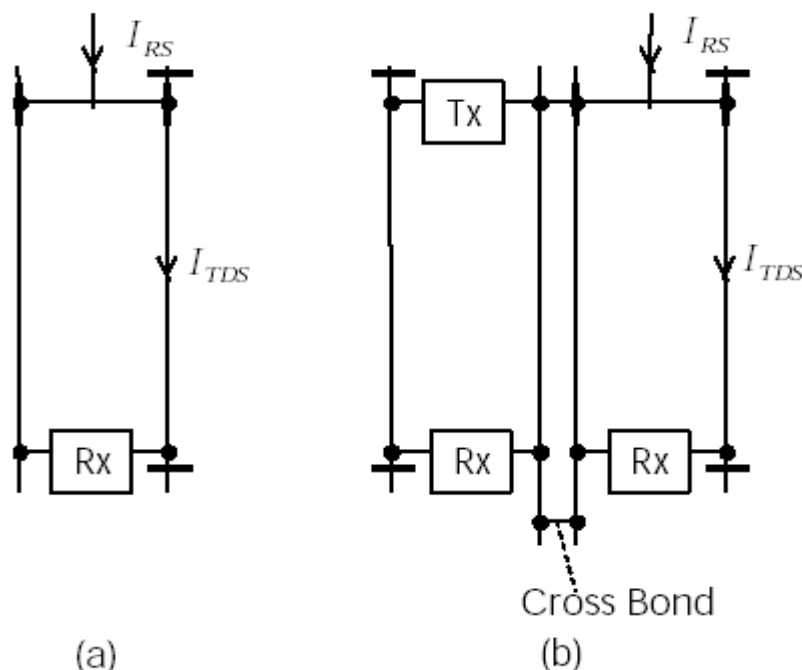


Figure 8: Infrastructure considered (worked example)

Using the parameters specified in section 3 the transfer function for the two cases can be determined.

For the 1200m track circuit of interest:

$$Z_{RAIL} = 1.2(346 \times 10^{-3} + j0.743 \times 10^{-3} \times 2\pi \times 372)\Omega$$

$$Z_{RAIL} = 0.415 + j2.084\Omega$$

$$Z_{RX} = 2.8\Omega$$

$$\Psi = \frac{Z_{RAIL}/n}{Z_{RAIL}/n + (Z_{RAIL} + Z_{RX})}$$

For $n = 1$

$$\Psi = \frac{0.415 + j2.084}{2(0.415 + j2.084) + 2.8}$$

$$|\Psi| = 0.384$$

For $n = 2$

$$\Psi = \frac{(0.415 + j2.084)/2}{(0.415 + j2.084)/2 + (0.415 + j2.084) + 2.8}$$

$$|\Psi| = 0.229$$

For the two cases shown in figure 7 these are as shown in table 1.

Infrastructure type	Transfer function (Ψ)
Case a - single track	0.384
Case b - double track	0.229

Table 1: Transfer functions for worked example

It should be noted that if the increased resistance (an additional 250 mΩ/km) of jointed track is used the value of the transfer function actually decreases, for case a it is 0.372 for case b it is 0.221.

5.2 Train current limits

If the transfer functions calculated in section 5.1 are applied to the permissible current in the primary of the track filter shown in figure 4 using the formula from section 4.3.1, then the curves of the maximum permissible interference current in the system can be found. These curves are shown in figure 9.

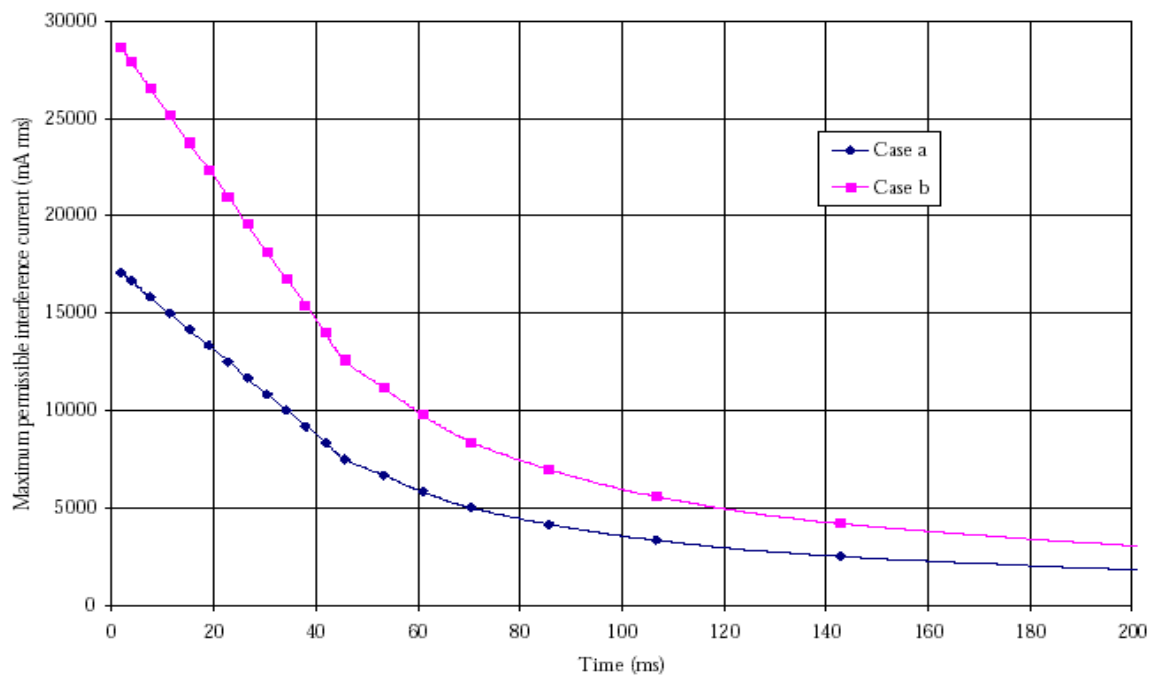


Figure 9: Total permissible interference current for the worked example <200ms

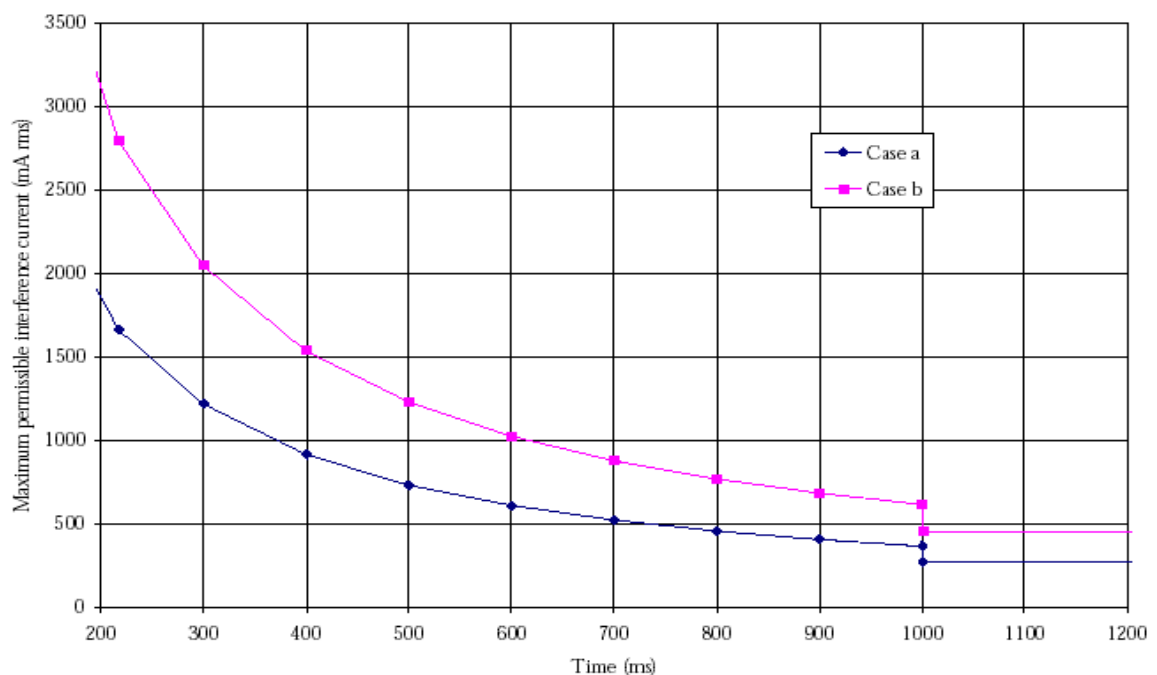


Figure 9: Total permissible interference current for the worked example >200ms

The limits for times exceeding 1 second (steady state) in figure 9 are 271mA rms for cases a and 454mA rms for case b.

The curves shown in figure 9 represent the total interference current that may exist in the traction supply system. Using a safety factor of three, as discussed in section 4.3.1, the permissible train current for the cases considered here are as shown in figure 10.

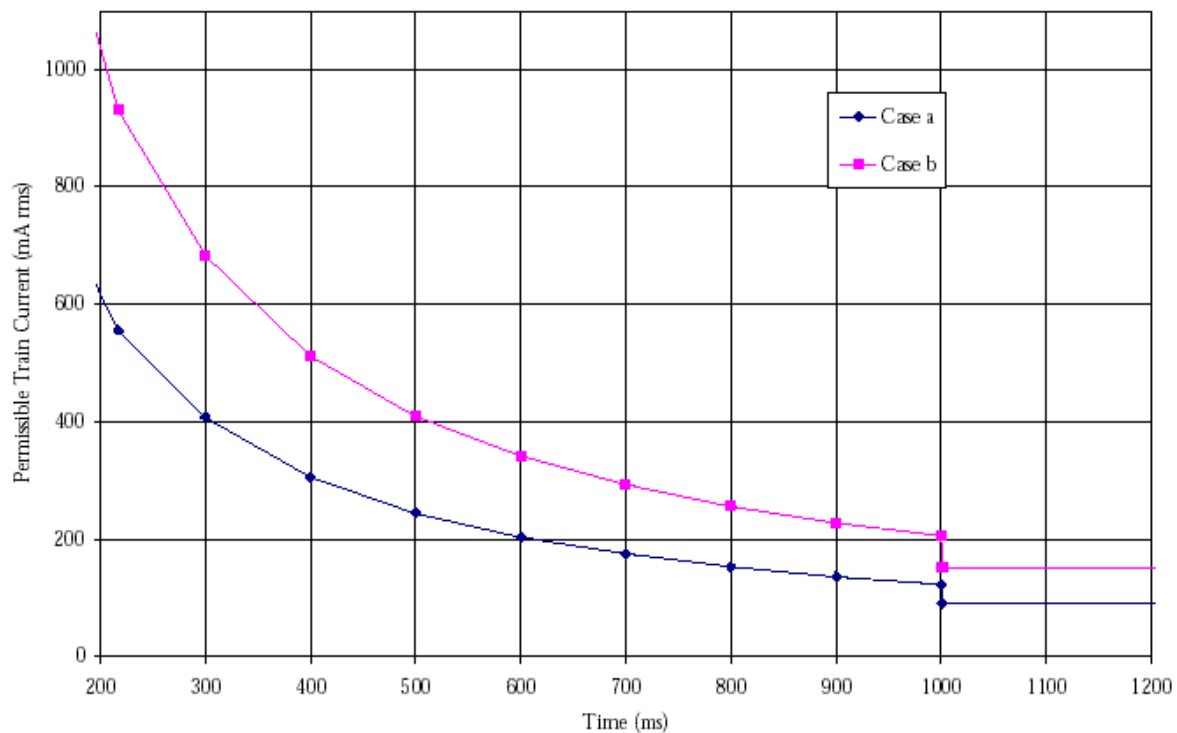


Figure 10: Permissible train interference current for the worked example >200ms

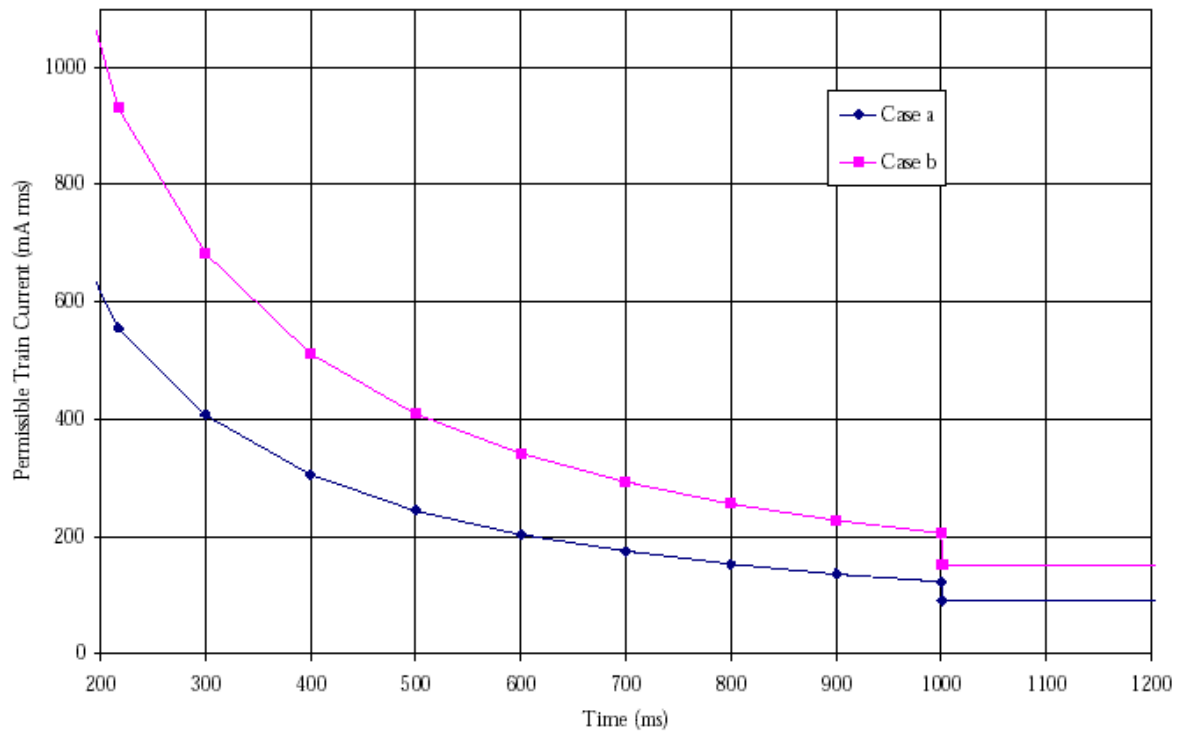


Figure 10: Permissible train interference current for the worked example >200ms

The limits for times exceeding 1 second (steady state) are 90.2mA rms for case a and 151 mA rms for case b.

6. INFRASTRUCTURE FAULTS

The analysis described in section 4 can be expanded to cover infrastructure fault scenarios by calculating a transfer function for the relevant scenario. A reasonable worst case scenario that can be considered is shown as case d in figure 11.

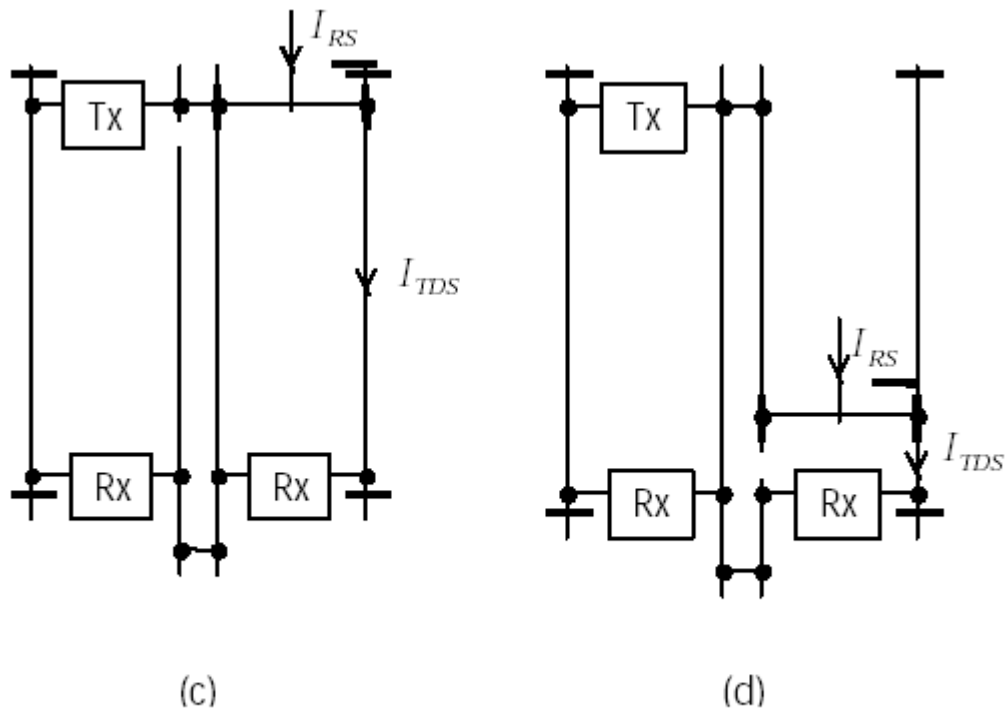


Figure 11: Fault scenarios for single rail reed track circuits

For case c shown in figure 11 the transfer function will be the same as that calculated for case a in section 5.1.1. (the single track case). For the worse case d shown in figure 9 the interference current is split between the path containing the receiver and section of signal rail and the path comprised by the traction rail of the occupied circuit in series with that of the adjacent track.

For case d a less pessimistic distance of 750m is assumed between cross bonds. The limiting case would be when the rail break is just before the receiver and the train is at the position shown, on the other side of the break.

In this case the impedance of the effective traction rail would be that of 1.5 km of rail therefore the transfer ratio can be calculated in the following way:

$$Z_{RAIL} = 1.5(346 \times 10^{-3} + j0.743 \times 10^{-3} \times 2\pi \times 372)\Omega$$

$$Z_{RAIL} = 0.519 + j2.605\Omega$$

$$Z_{RX} = 2.8\Omega$$

$$\Psi = \frac{Z_{RAIL}}{(Z_{RAIL} + Z_{RX})}$$

$$\Psi = \frac{0.519 + j2.605}{0.519 + j2.605 + 2.8}$$

$$|\Psi| = 0.63$$

This leads to the calculation of a transfer function, ψ , of 0.63. This factor is highly pessimistic due to the length of the train and the signal rail of the adjacent track being ignored. If the length of the train was considered this would shorten the length of the traction rail return path thus reducing it's impedance and in turn the transfer function. With such a configuration the signal rail of the adjacent track may, in fact, carry some interference current and further reduce the transfer function.

In the case of an infrastructure fault a reduced safety factor of 2 can be applied due to the following considerations:

- The fault will exist only for a limited time.
- The pessimisms already encompassed in the factor of 3 described in section 4.3.1 also apply.
- The length of the train has been ignored, including this would result in a significant reduction in the value of ψ .

Applying the value for Ψ of 0.63 and safety margin of 2 leads to a permissible train current curve shown in figure 12. The limit for times exceeding 1 second (steady state) this case is 82.5mA rms.

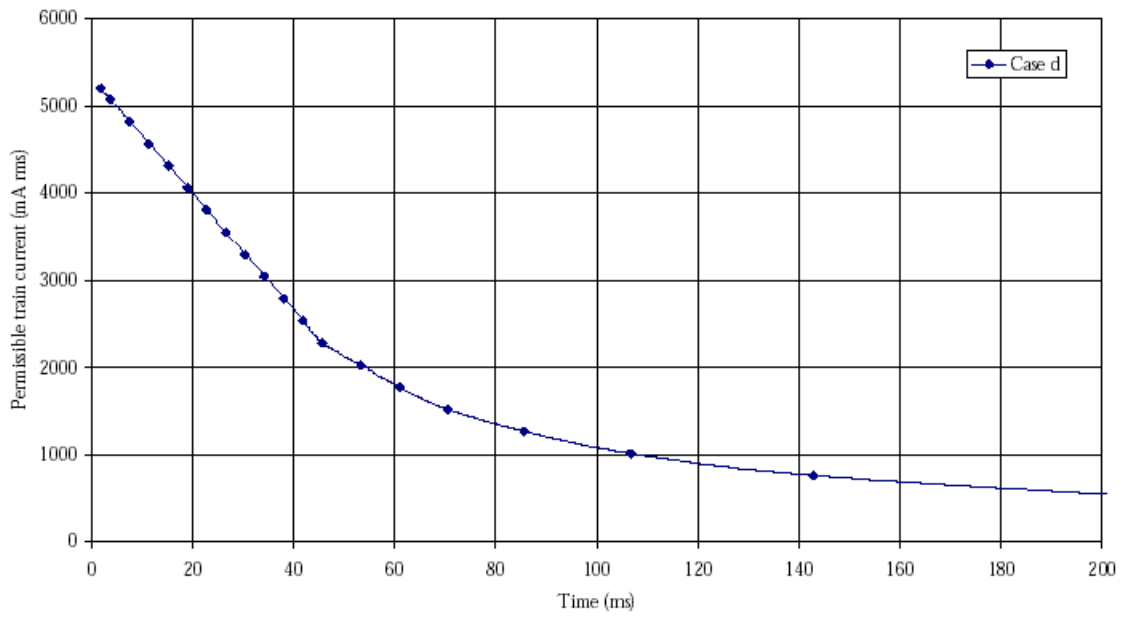


Figure 12: permissible train current under infrastructure fault condition case d shown in figure 9 <200ms

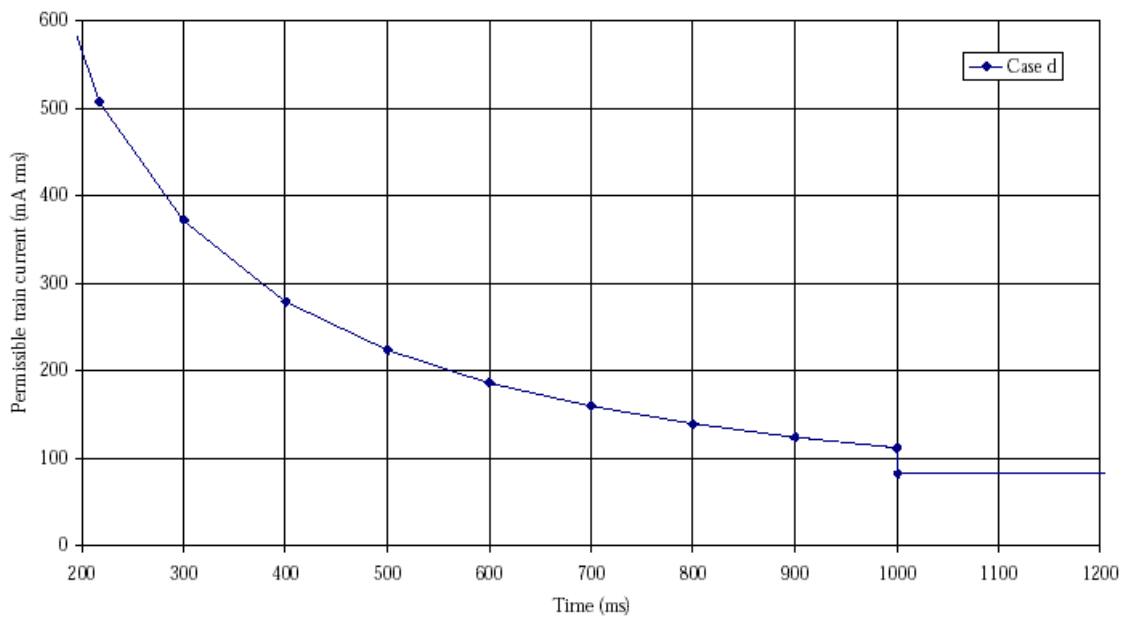


Figure 12: permissible train current under infrastructure fault condition case d shown in figure 9 >200ms

7. METHODOLOGY FOR DEMONSTRATION OF COMPLIANCE

The longest track circuit on the route of interest must be determined by recourse to the RAR (Railtrack Asset Register) or other source. It is expected that this will be shorter than the 1200m track circuit analysed in section 5.

It must be confirmed that the track circuits are installed in single rail mode only.

It must be confirmed that track circuits are of the RT type or if R type the receivers are not fitted with step-up transformers. Since the analysis is based on the most sensitive receiver setting there is no requirement to place any restrictions on future adjustment of reed track circuit receivers on the route except the fitting of step up transformers which must not be allowed.

The rail-break figures and times at risk are indicative only and should be replaced by the latest available data and/or statements from Railtrack.

7.1 Normal train - normal infrastructure

Once the longest track circuit on the route is determined the procedure shown in sections 4 and 5 can be followed to evaluate the worst case susceptibility for the track circuits on the route for the case of normal infrastructure. This is then reduced by the safety margin of 3 to determine the train current limits. For the 1200m track circuit examined in section 5 of this document the steady state limit is 90.6mA rms. This can then be compared to the train emission data. Under normal, including transient, operating conditions this limit must not be exceeded.

7.2 Faulty train - normal infrastructure

From the failure analysis carried out the rate at which the train exceed the curves calculated for normal infrastructure in section 8.1 should be determined. The frequency of occurrence, duration and magnitude of the exceedance should be taken into account when constructing mitigating arguments.

7.3 Normal train - faulty infrastructure

The faulty infrastructure scenario as described in section 6 is used to determine a permissible time current curve for a given fault scenario. This can then be apportioned down by a safety factor to determine a train current limit. In the case of faulty infrastructure a reduced safety margin can be applied, in this case a factor of 2

can be used. This train current limit so calculated is 82.5 mA rms. This can then be compared to the train data.

In this case there may be specific normal transient events that may exceed the permissible current curve. These events should be quantified in terms of the rate at which they occur and the overall rate of potential WSFs is determined by combining them with track failure data.

This can be done in the following manner.

For the case of a broken rail occurring on a particular track circuit. The total number of rail breaks per year is 680 (ref. I 6) per 20,000 track miles and the time at risk is 12 hours taken as the average time for a rail break to be discovered (ref. I 3). The number of railbreaks per year for the section of railway of interest must be confirmed by recourse to latest available figures or statements available from Railtrack. The probability of a track circuit having a broken traction rail at any one time is:

Probability of track circuit having broken rail (P_{tcbr}) =

No of breaks/track miles * TC length(miles)*(time at risk/no of hours per year)

The predicted rate for a train which is standing on a track circuit exceeding the susceptibility level and the traction rail being broken is therefore:

$P_{tcbr} * \text{Exceedance rate (per hour)}$

This will then give the failure rate in exceedances per hour for normal train and failed infrastructure.

7.4 Faulty train - faulty infrastructure

In this case the train limits and the probability of a track circuit having a broken rail (P_{tcbr}) are determined in the same way as in section 8.3. The exceedance rate for train failures is predicted by the failure analysis. This failure rate is then used in conjunction with the probability of a track circuit having a broken rail to determine the overall exceedance rate for this case.

7.5 Safety targets

The overall exceedance rate shall be found by adding the rates calculated in sections 7.2, 7.3 and 7.4. This total exceedance rate shall be shown to be ALARP in the context of the safety requirements of the project.

7.6 Axle to axle voltages

The axle to axle voltages should be compliant with the limit defined in section 4.3.2.

8 Appendixes

APPENDIX I: PARAMETERS

Parameter values used, assumed variation limits, sensitivity to variation and supporting references

Parameter:	Rail impedance
Value:	0.346 Ω /km 0.743mH/km
Expected variation:	The expected variation over the range of reed track circuits (363 - 423 Hz) is in the order of 1%.
Source:	"Track parameters for interference studies" BRR report number: RR: STG-D-EPS(373)-RP-18, Issue 1, dated 19th January 1995.

Parameter:	Reed receiver 'pick-up' criteria
Value:	88000 mV*ms, between 200ms and 1 second, based on further test results below 200ms, drop away criteria 65mV used for durations above 1 second.
Expected variation:	These limits are expected to be applicable for all RT type reed track circuit receivers. The 'pick-up' criteria is based on the most sensitive filter from a sample tested
Source:	"Determination of Permissible Rail/Wheel Interface Current Levels for Single Rail Reed Track Circuits on the AC Railway", AEAT Rail ref. 120-dr0397, version b0, dated May 1999. "Reed Receiver Test Report - Railtrack Industry Data Initiative", AEAT Rail, ESID9079, version 2, March 2000.

Parameter:	Ratio of input current to output voltage for track filter
Value:	1.6 mA/mV
Expected variation:	This value is independent of the setting of the filter and is expected to be applicable for all RT type reed track circuit receivers
Source:	"Determination of Permissible Rail/Wheel Interface Current Levels for Single Rail Reed Track Circuits on the AC Railway", AEAT Rail ref. 120-dr0397, version b0, dated May 1999.

Parameter:	Track filter primary impedance at resonance
Value:	2.8 Ω resistive
Expected variation:	This value is based on the track filter of an RT type reed receiver at its most sensitive setting and is expected to be applicable for all RT type reed track circuit receivers
Source:	"Determination of Permissible Rail/Wheel Interface Current Levels for Single Rail Reed Track Circuits on the AC Railway", AEAT Rail ref. 120-dr0397, version b0, dated May 1999. "Reed Track circuit parameters" fax from D. M. Poole to N Rambukwella dated 12/5/93.

APPENDIX 2: ASSUMPTIONS

Configuration, supporting references and infrastructure failure modes

General

Transients are not repetitive (analysis considers response to single transient only, with no reset time being defined).

A broken signal rail will lead to a protective RSF.

The failures analysed are limited to a broken traction rail, or a broken cross bond (equating to the single track case) in a two track scenario.

Results from tests conducted on receiver equipment of one operating frequency can be applied to other operating frequencies.

There are no redundant IRJs in the track circuit (which would increase the effective rail break rate).

Configuration

The track circuits are assumed to be installed in single rail mode with no impedance bonds present.

Double rail track circuits with impedance bonds require individual analysis. Type R track circuit receivers are assumed not to have step up transformers fitted.

The effect of the connection of the OHLE masts to the rail is ignored. This is a pessimistic assumption.

All of the return current flows in the rails - there is no leakage between the rails. This is a pessimistic assumption.

The effect of booster transformers is ignored i.e. rail return only. This is a pessimistic assumption.

Cross bond spacing is assumed to be 750m. This is only really significant for the worst case fault condition and is considered to be applicable without major exception as this is given in Railtrack standards.

Infrastructure failure modes

It has been assumed that only one infrastructure failure can occur on one track circuit at one time.

In the case of reed track circuits a cross bond failure is equivalent to the single track railway case.

The broken rail scenario examined considers a broken rail at the worst position with regard to its position relative to receiver and train location. The length of the train is ignored, in reality the length of the traction return path would be reduced by the train length leading to less current flow through the track filter.

APPENDIX 3: HAZARDS

Interference current in traction return current

This hazard is covered by the assessment as presented.

Axle to Axle or Longitudinal voltage

This hazard is covered by the assessment as presented.

Direct induction into track mounted or trackside components

Direct induction has been considered for the Class 373/2 safety case where the line current required to cause maloperation of reed track circuits is shown to be higher than the limits for traction current flowing through the primary of the track filter calculated in this document. (ref HR/EUKL/NLL/IO/4 “Reed track circuit safety assessment”).

Particular transient events

No single type or source of transient has been identified as posing a particular risk to reed track circuits. All transients may contain reed frequency interference therefore train testing must include all foreseeable transient conditions.

Wheelset impedance

The maximum wheelset impedance is not quoted in this document

Vehicle geometry (overhang, wheelbase etc..)

This hazard is not covered by this document.

RSF due to excessive off channel current

This hazard is not covered by this document.

APPENDIX 4: ASSUMPTIONS

Pessimistic assumptions

Zero train length for line current limits.

Train length equal to cross bond spacing for longitudinal voltage limit.

Rail return only for traction current.

All of the return current flows in the rails.

Exceedance of permitted current curve may only be transient in nature, leading to a short duration, time limited WSF.

A higher permissible level would be applicable to the High Performance variant.

A worst case train position is considered in the analysis;

No TPR is assumed this does not have a significant effect on the analysis or permissible train current.

Assumptions/variations covered by safety margin

Normal variation of track circuit receiver performance.

Background/coincident sources of interference covered by safety margin

Other rolling stock.

Transmitter breakthrough.

APPENDIX 5: SOURCES

Sources

The technical work contained in this document is taken from the Class 92 EMC safety case, produced by AEA Technology Rail, ISAs were WS Atkins Rail (secretariat reviewer) and Mott MacDonald.

Characterisation of reed receiver components has been cross checked against testing carried out for the ALSTOM 'Juniper' trains. This work is also the source of the drop away criteria and the axle to axle voltage limit methodology. This work was undertaken by AEA Technology Rail, ISA was WS Atkins.

RAILTRACK

BRIEFING NOTE

RT/E/C/50001 to RT/E/C/50019

Issue 1

Methodology for the demonstration of electrical compatibility between rolling stock and infrastructure

Issue Date: February 2003

Background / Synopsis

This suite of Codes of Practice is intended to assist train-builders, TOCs, and ROSCOs in the production of Route Acceptance Safety Cases for electric trains, by describing a methodology for demonstrating compatibility with EE&CS systems on Railtrack's electrified routes.

The electrical susceptibility of the key items of infrastructure is described and characterised, and recommended methods for demonstrating compatibility between rolling stock and infrastructure are given.

Key Changes / Compliance

This is a new suite of documents. As the documents are Codes of Practice, compliance is not mandatory.

Implementation

The information contained in the documents may be used with immediate effect. The various documents in the suite will be published progressively during 2003.

Of Interest To:

Authors of Route Acceptance Safety Cases.

Name and Title of Author

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